Simulation and Modeling

Lecture 6

Inventory System Analysis

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Computational Model

Computational Model: Simulator

Member variables

- Clock_
- eventList_

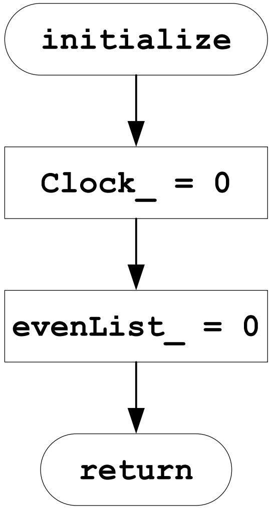
Member Functions

- Initialize ()
- Run ()
- addEvent (Event *e)
- Event* removeEvent ()

Computational Model: Simulator

initialize () function

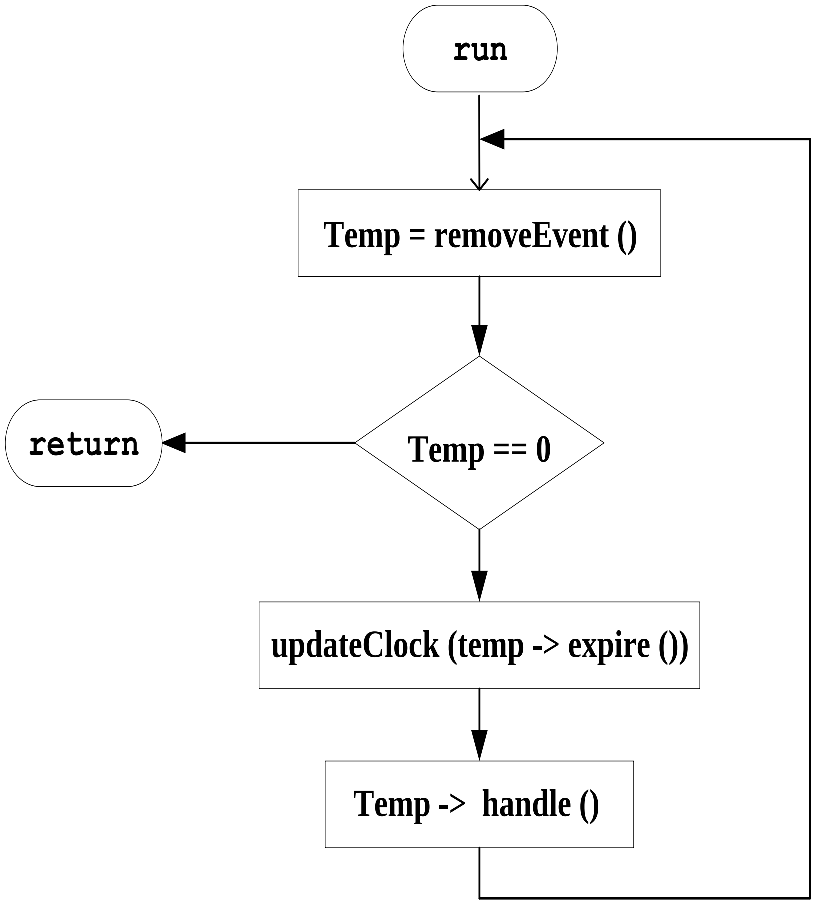
- Initialize the simulator
- Set the clock_ to 0
- Set the eventList_ to null



Computational Model: Simulator

run () function

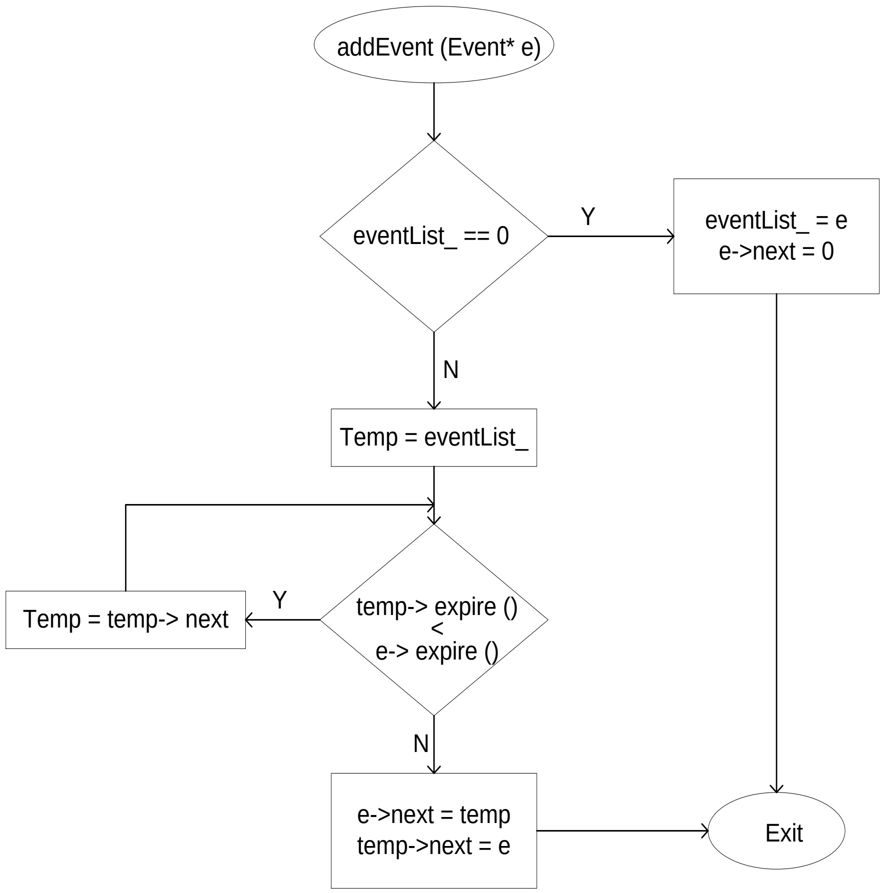
- Schedule the events
  - First event in the eventList_ is scheduled
  - If there is no event in eventList_, simulation is terminated
- Update the simulation time clock_ to the current time
- Call the event handling function
- Move to the next event in the list



Computational Model: Simulator

addEvent () function

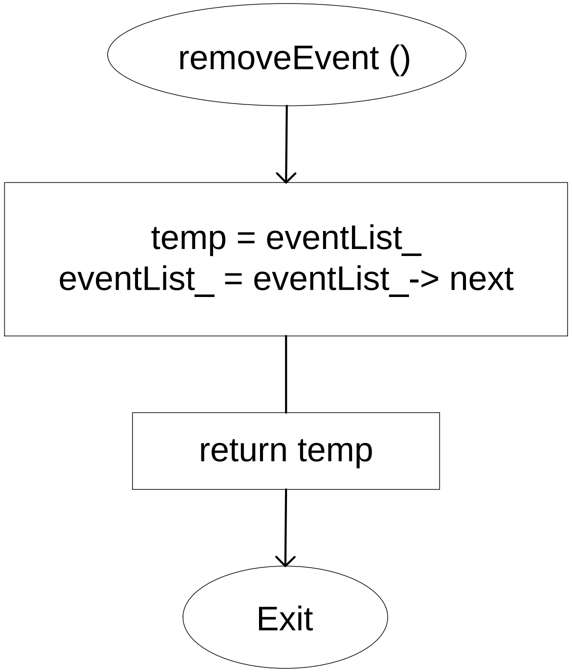
- Add an event in the eventList_
- Check whether the event is added
  - in an empty list
  - as first element in the list
  - in the middle of the list
  - as last element of the list
- Flow chart shows two cases
  - Empty list
  - Middle of a list



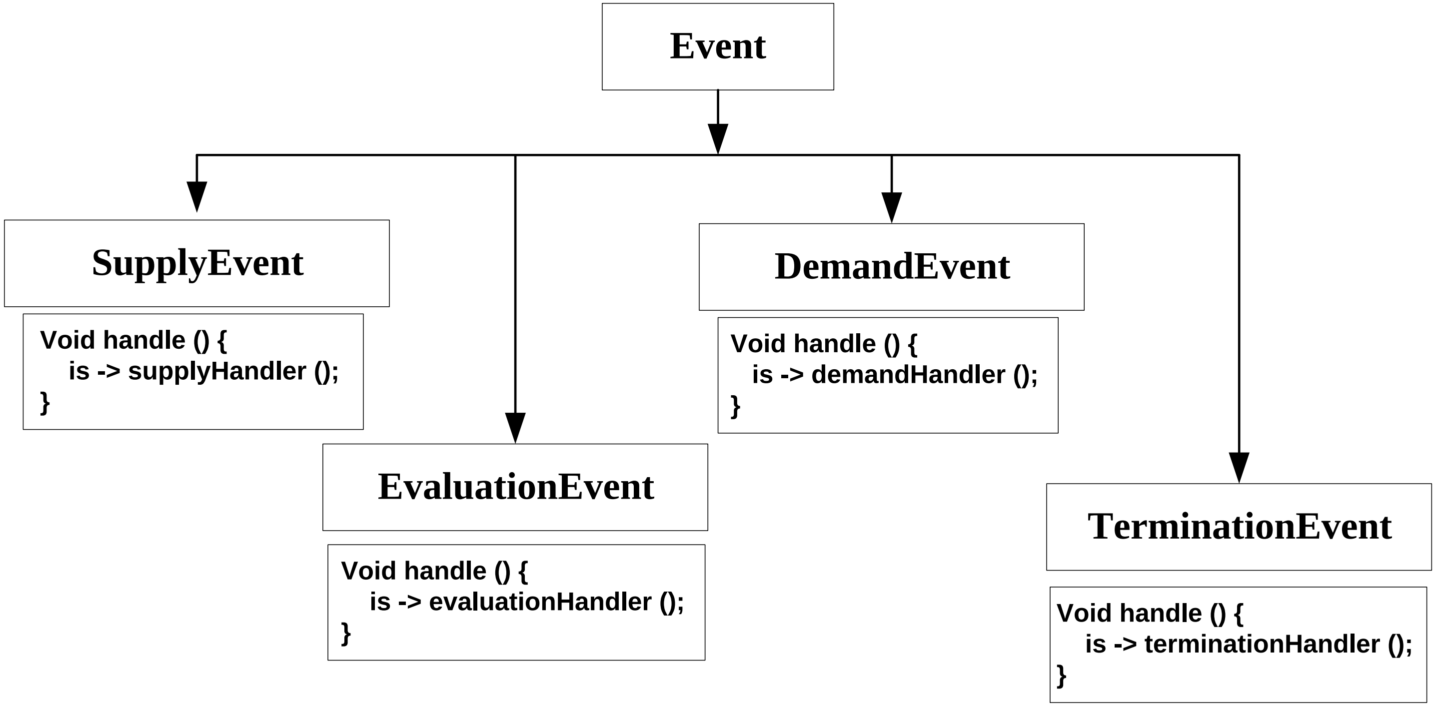
Computational Model: Simulator

removeEvent () function

- Remove the first event of the evenList_
  - Return the event
- If the list is empty returns null



Computational Model: Events



Computational Model: Inventory

Member variables

- State variables
  - ivLevel_
- Events
  - d_, e_, s_, t_
- Constants
  - maxLevel_, minLevel_, n
  - minLag_, maxLag_
  - setupCost_, itemCost_
  - arrivalMin_, demandMin_
- Statistical Variables
  - areaHolding_, areaShortage_
  - orderingCost_
- Temporary Counters
  - orderAmount_
  - timeLastEvent_
  - evalId_, arid_, demndId_
  - itemOrdered_

Member Functions

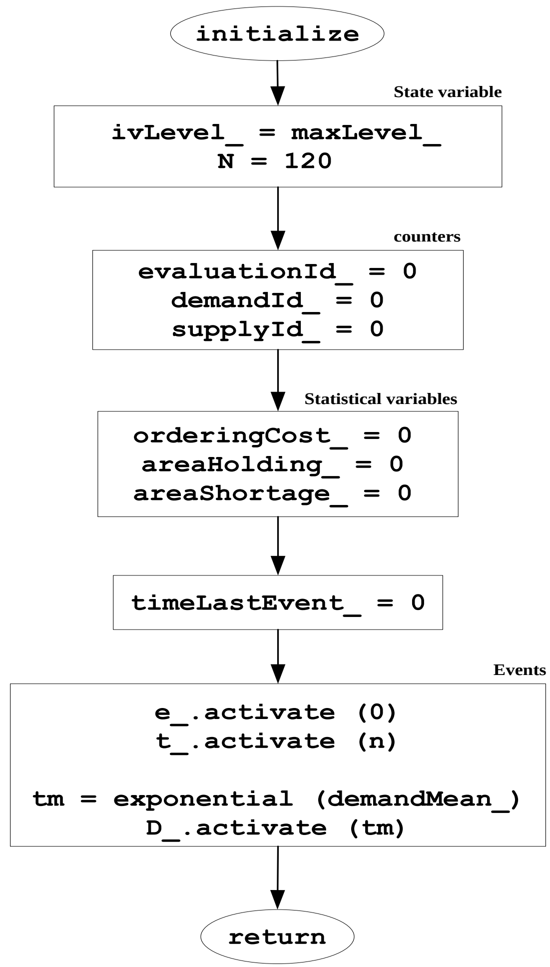
- Initialize ()
- demandHandler()
- evaluationHandler()
- supplyHandler()
- terminationHandler()

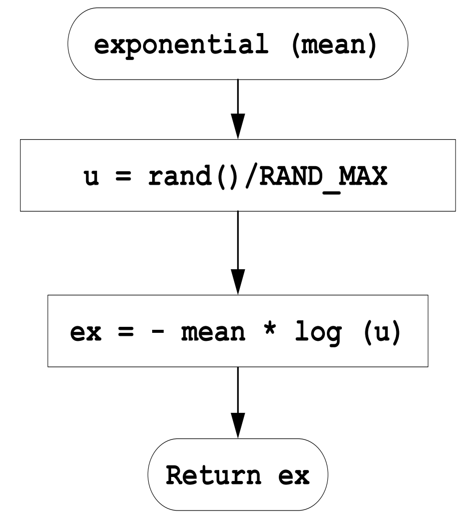
- Exponential (mean)
- uniformRand(min, max)
- discreteRand()

Computational Model: Inventory

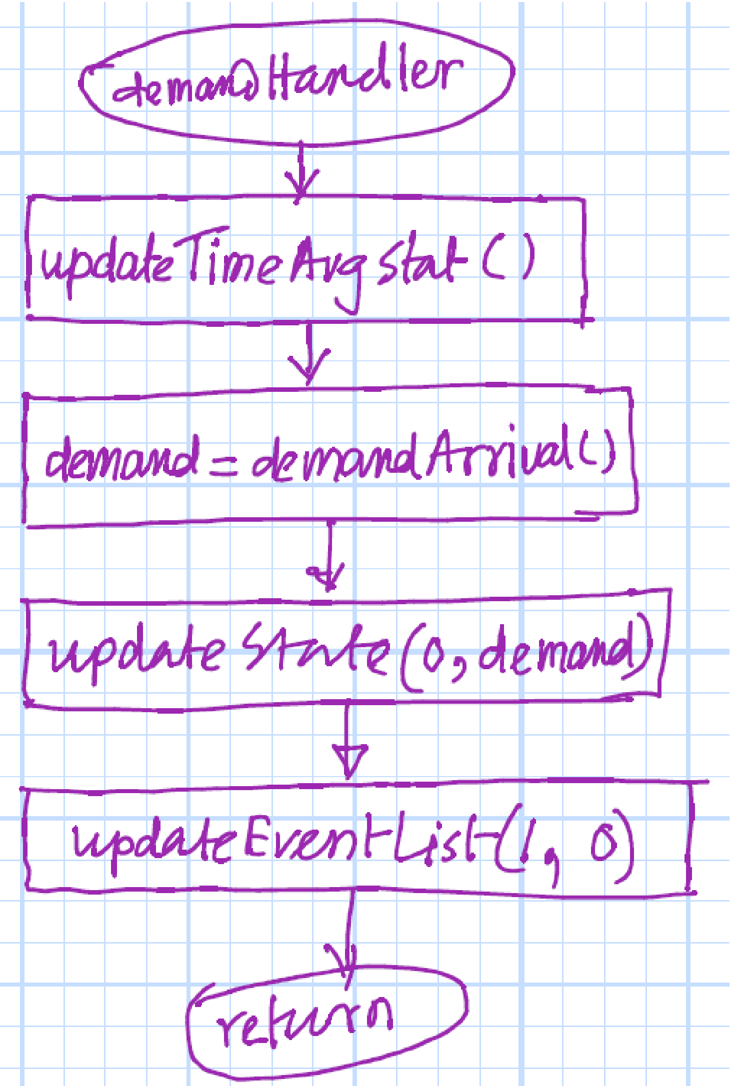
<u>initialize () function</u>

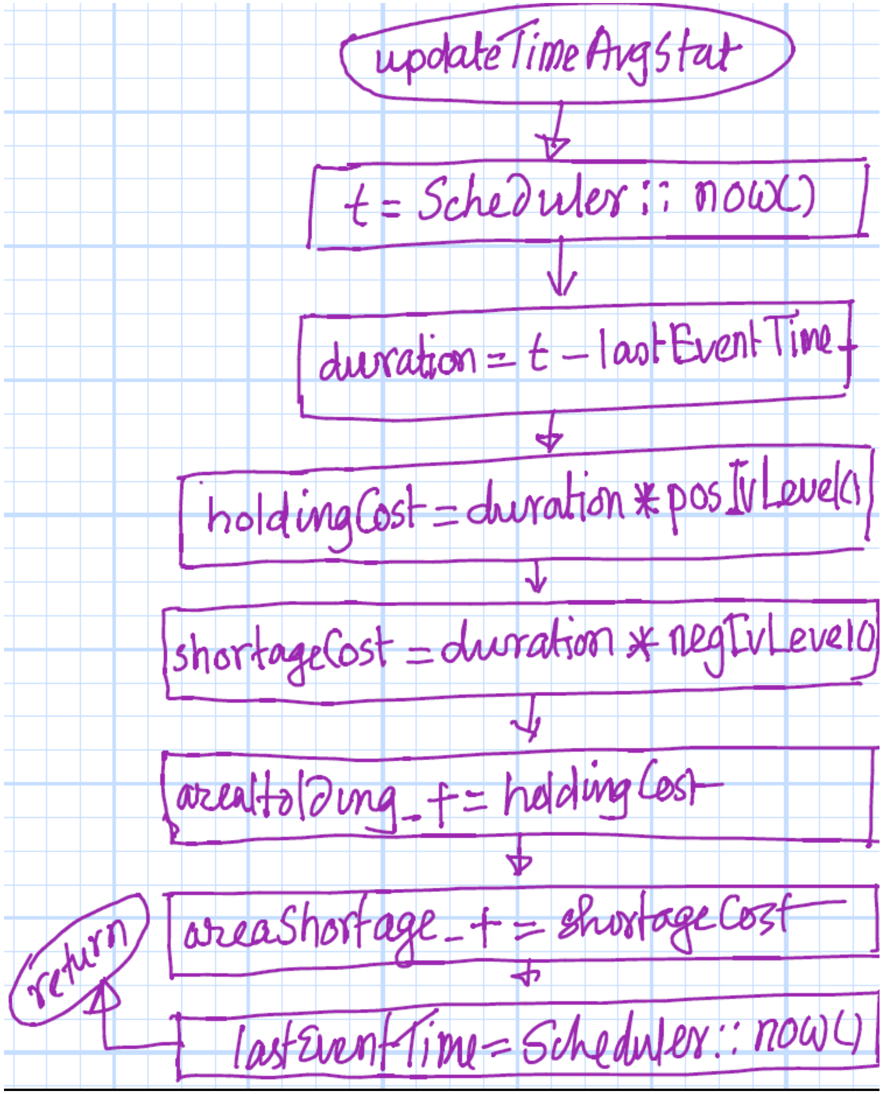
- Initialize the state variables
  - ivLevel_ = 0
- Initialize input variables
  - n = 120
  - demandMean_ = 0.1
  - $S_L = 20, S_M = 40$
  - minLag_ = 0.5, maxLag_ = 1.0
- Initialize counter and output variables
- Finally, activate evaluation and demand events



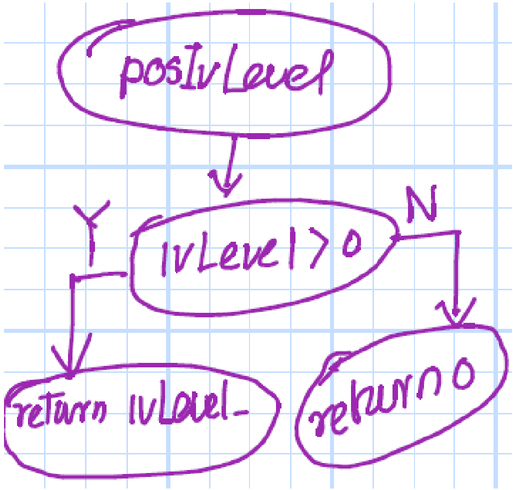


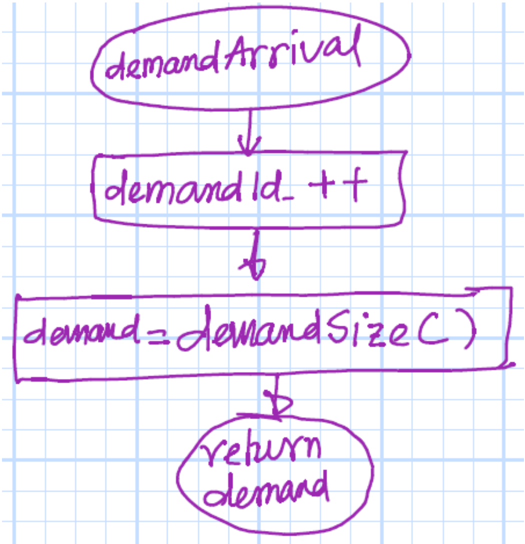
IS: Demand Handler

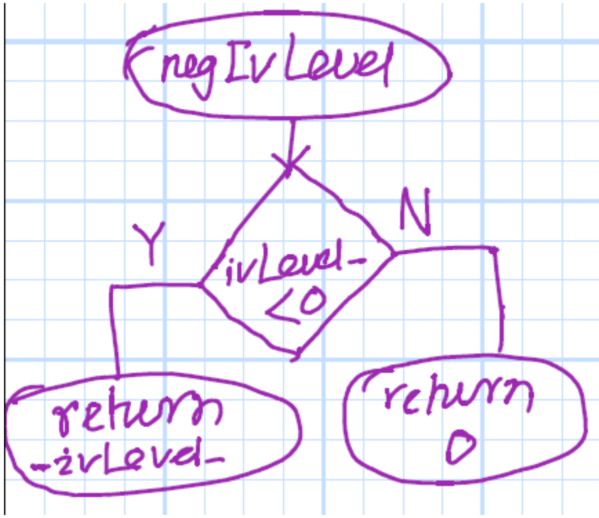


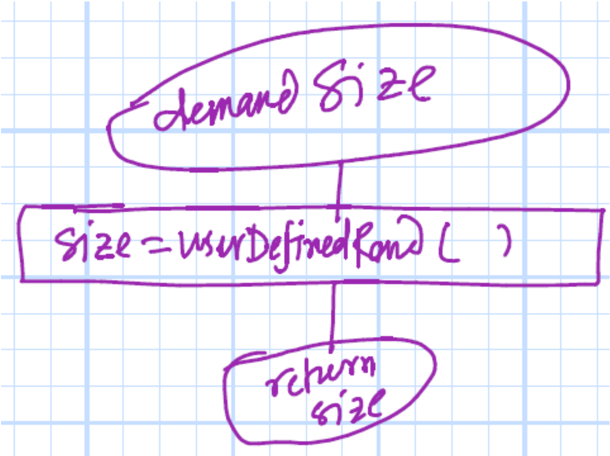


IS: Demand Handler

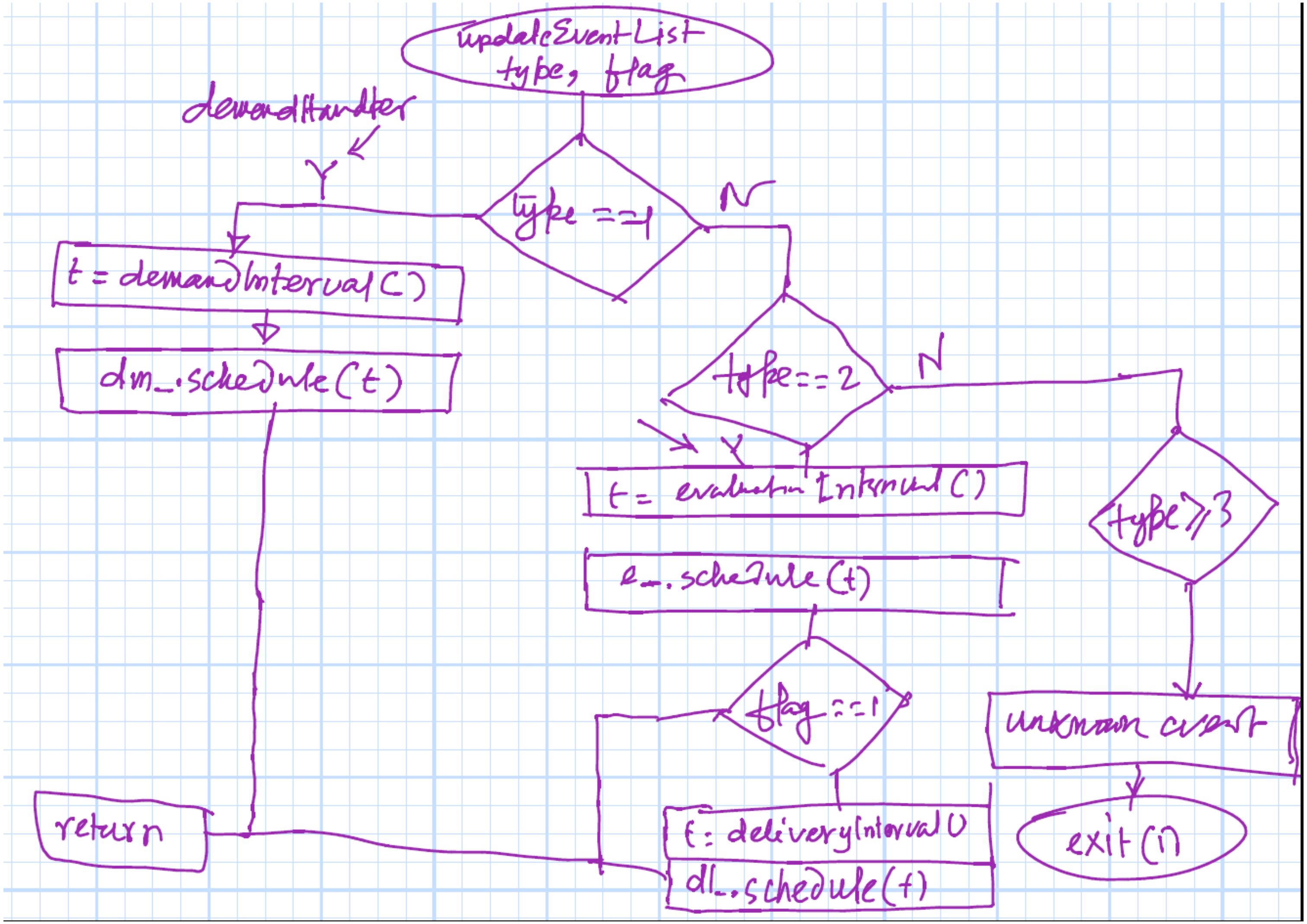




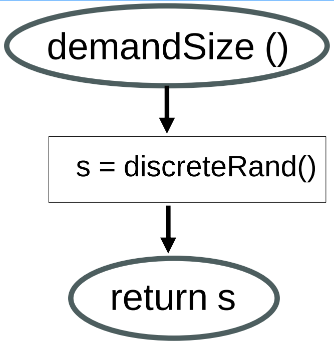


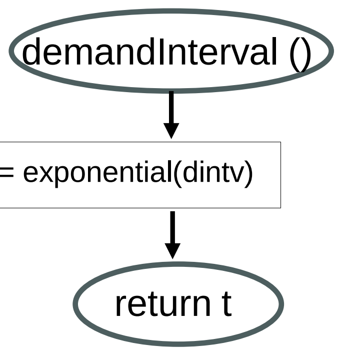


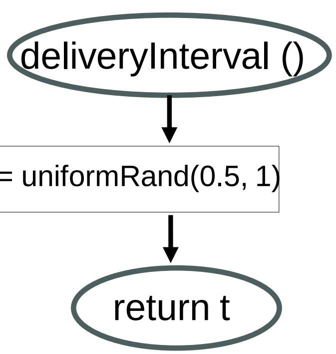
IS: Demand Handler

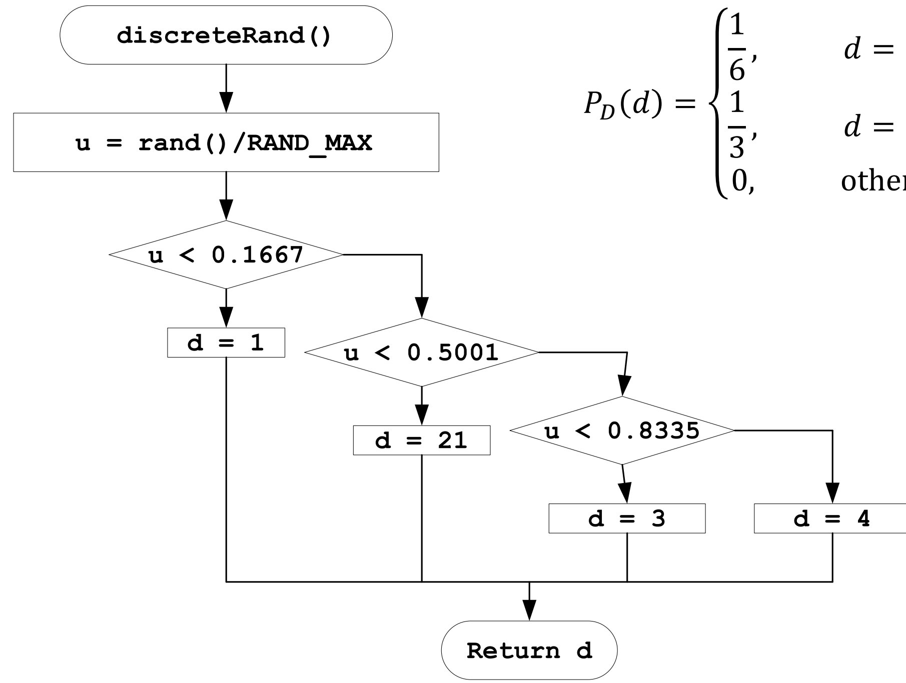


IS: Demand Handler

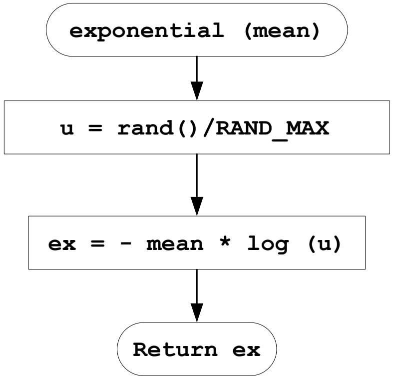


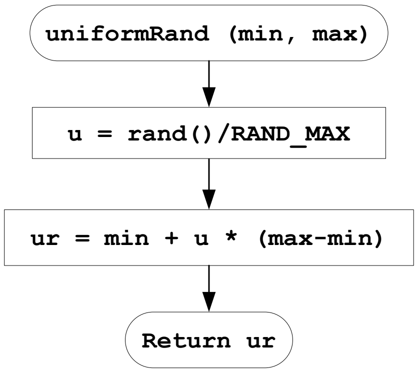




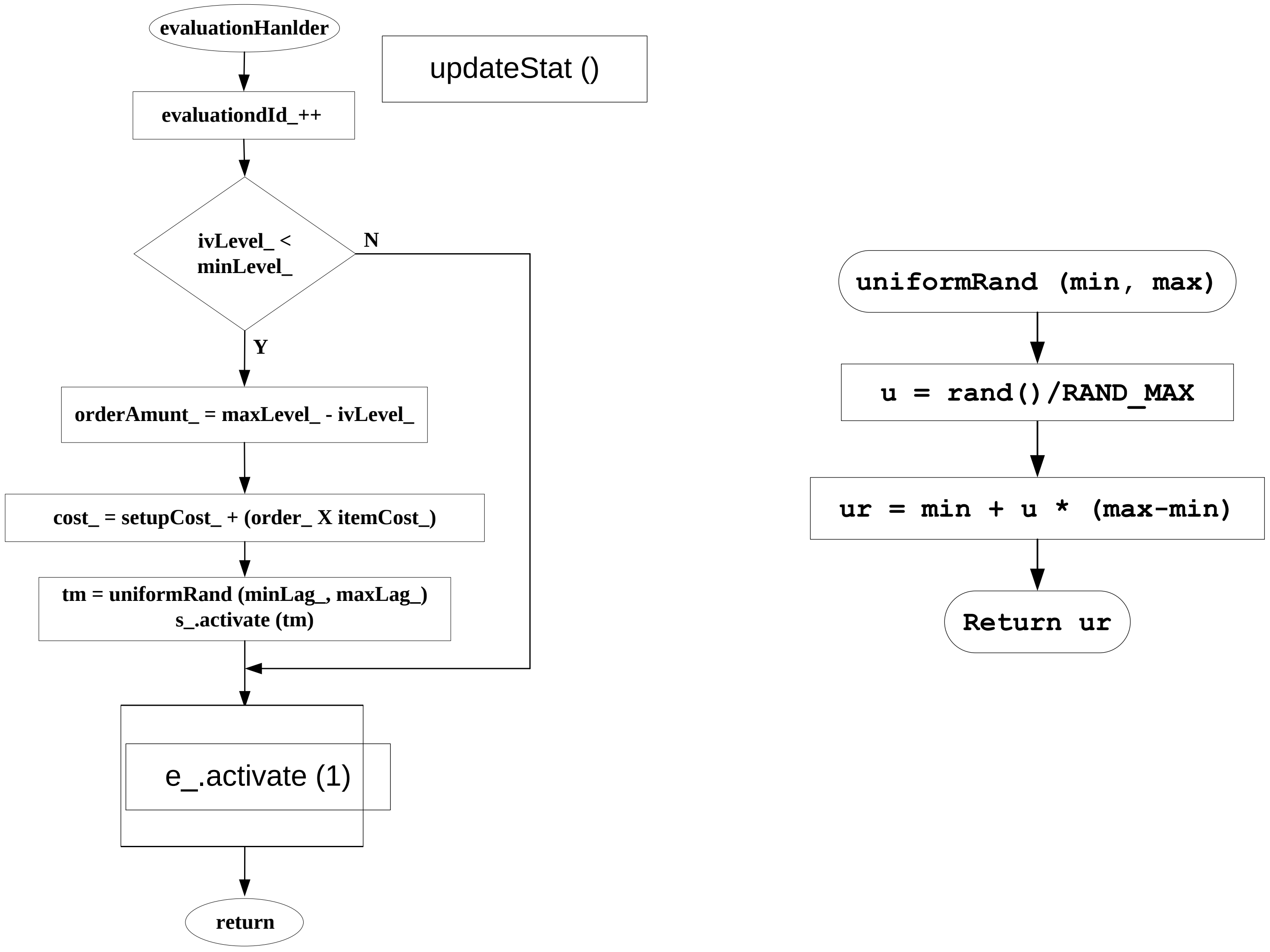


$$P_D(d) = \begin{cases} \frac{1}{6}, & d = 1, 4 \\ \frac{1}{3}, & d = 2, 3 \\ 0, & \text{otherwise} \end{cases}$$

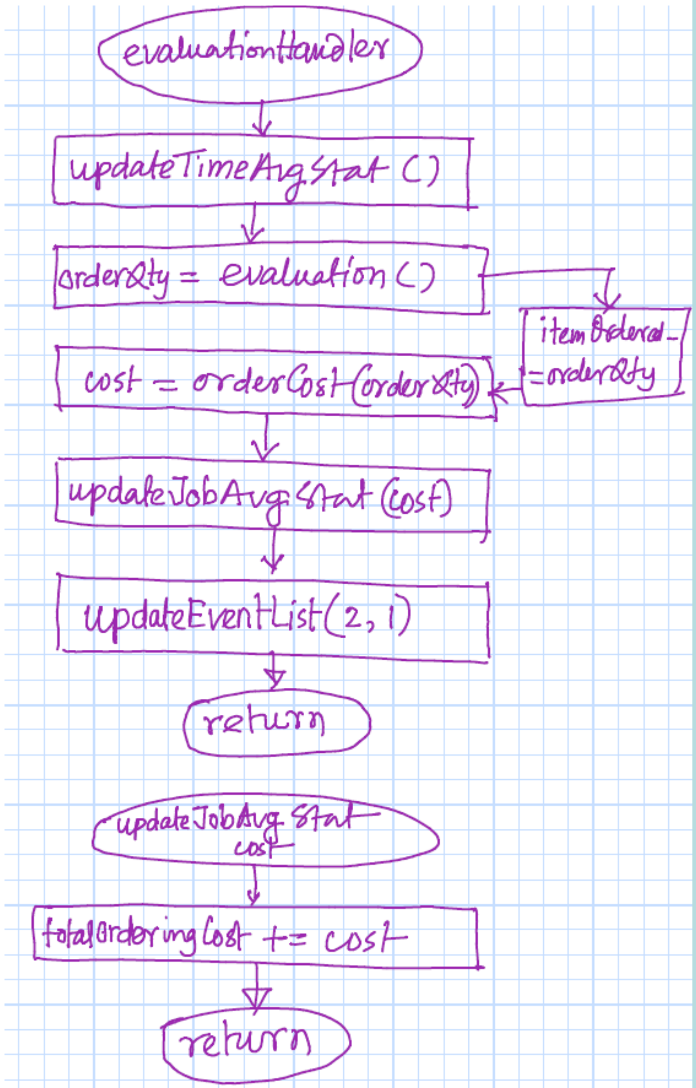


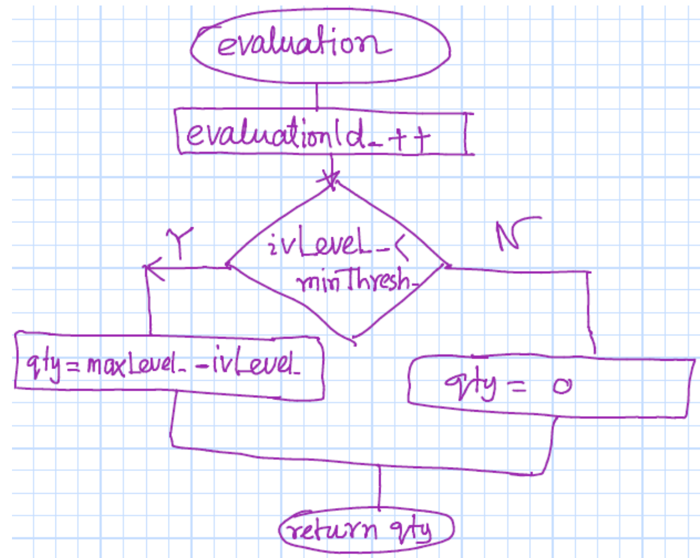


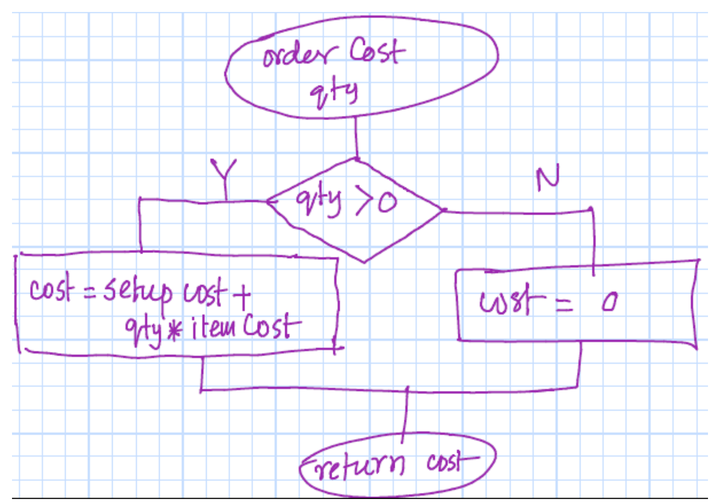
IS: Evaluation Handler



IS: Evaluation Handler

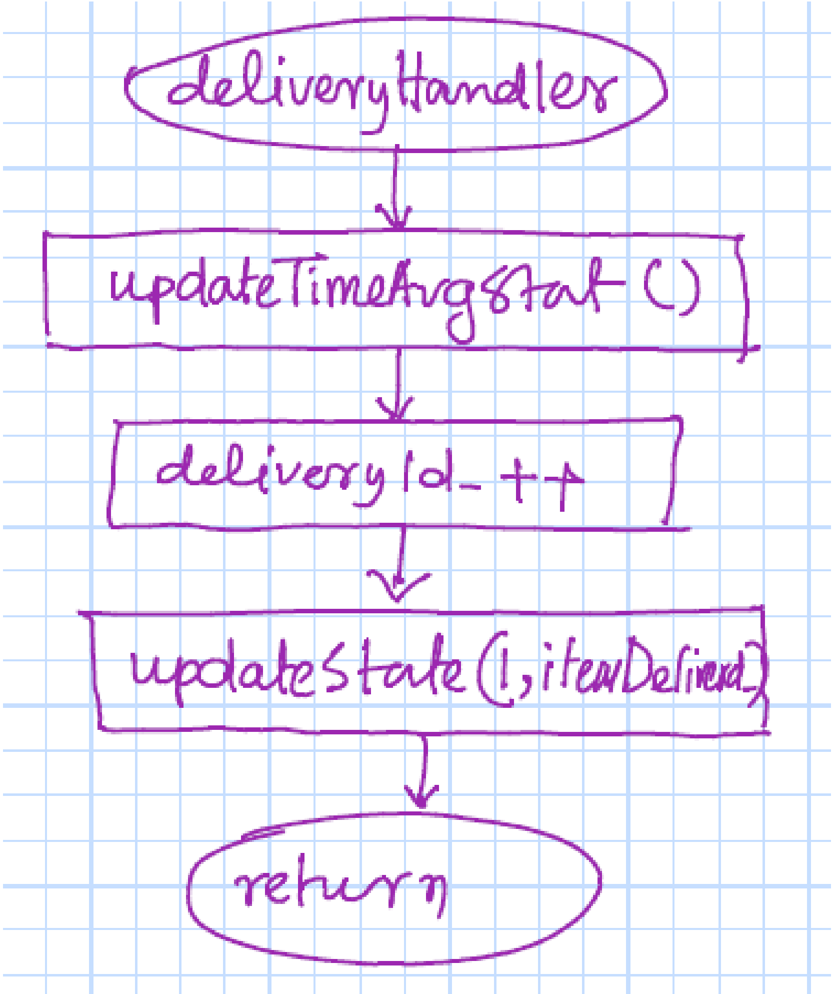






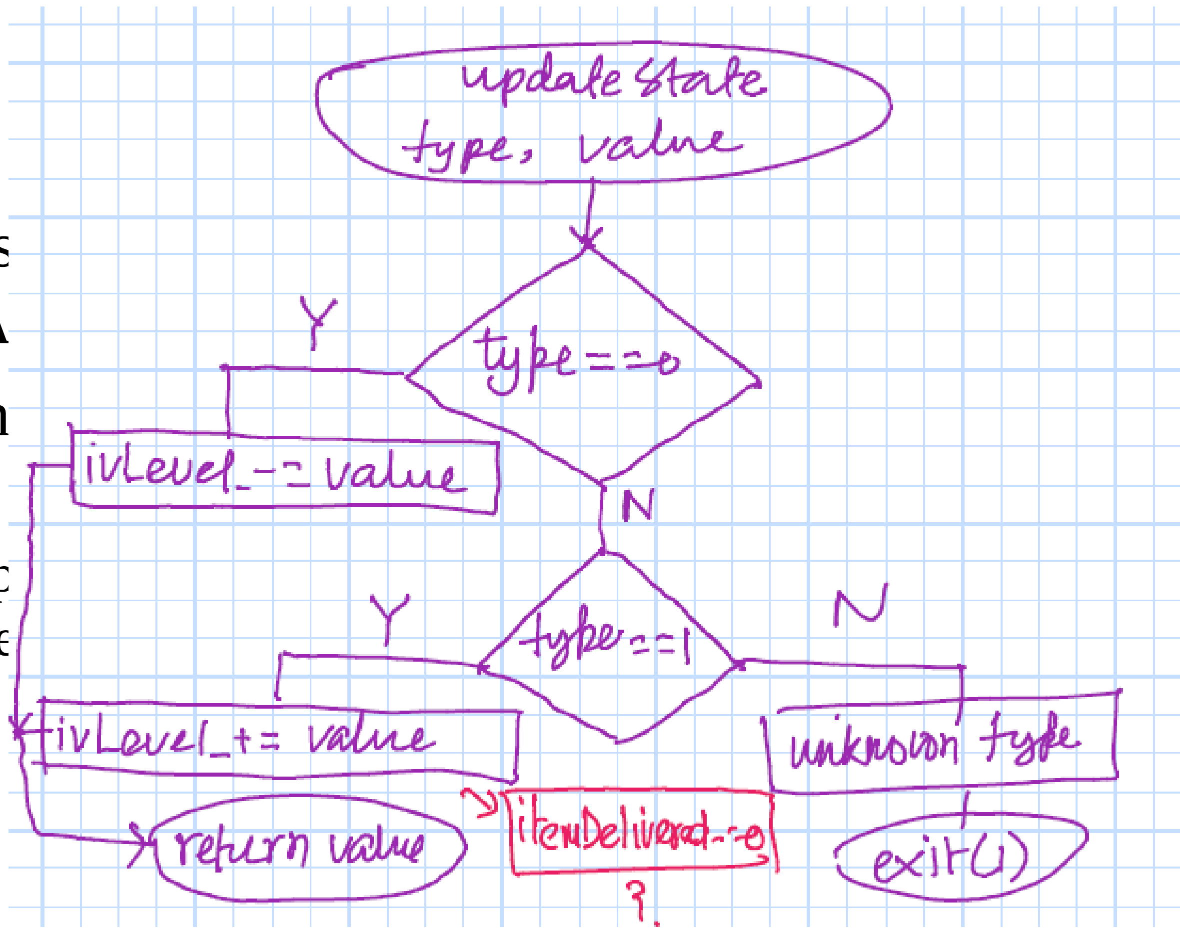
IS: Delivery Handler

- Increase the ivLevel_ by the orderAmount_
- Does not schedule a supply event
  - supplyEvent is schedule by the evaluation event



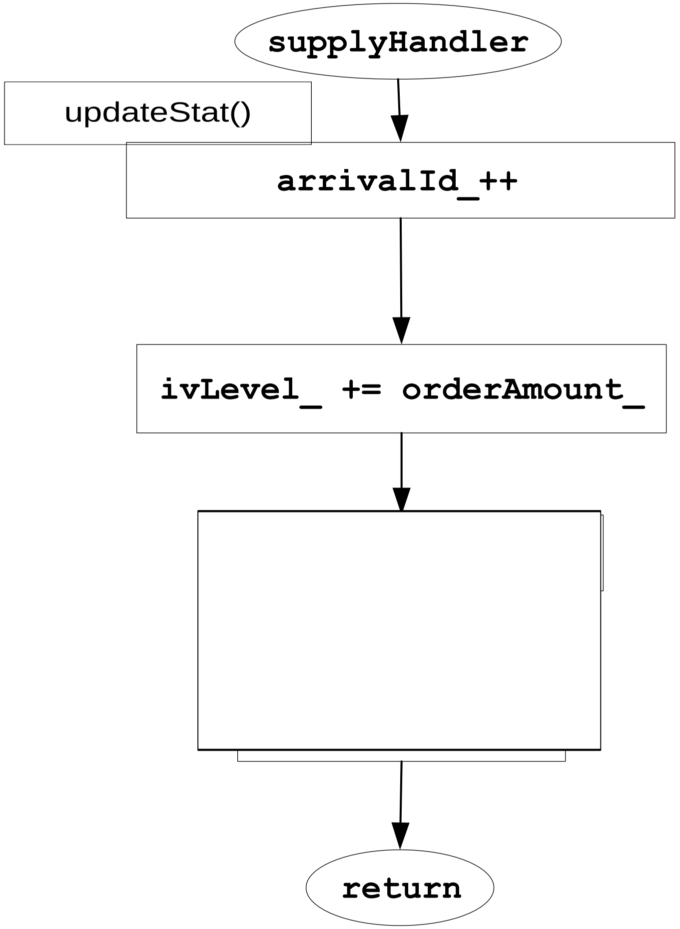
IS: Delivery Handler

- Increas
  orderA
- Does n
  event
  - supp
    the e



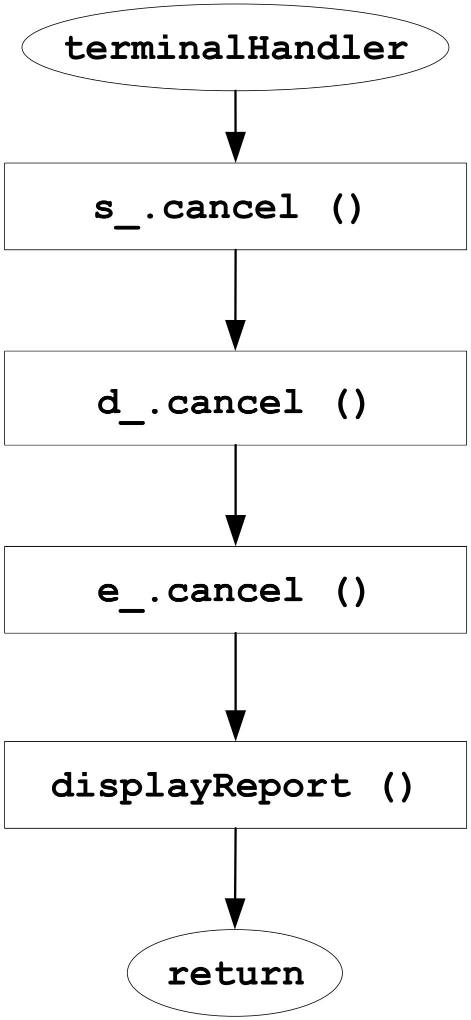
IS: Supply Handler

- Increase the ivLevel_ by the orderAmount_
- Does not schedule a supply event
  - supplyEvent is schedule by the evaluation event



IS: Termination Handler

- Remove all the event from the evenList_
- Next iteration of the run function will remove a null from the evenList_
- Terminate the simulation



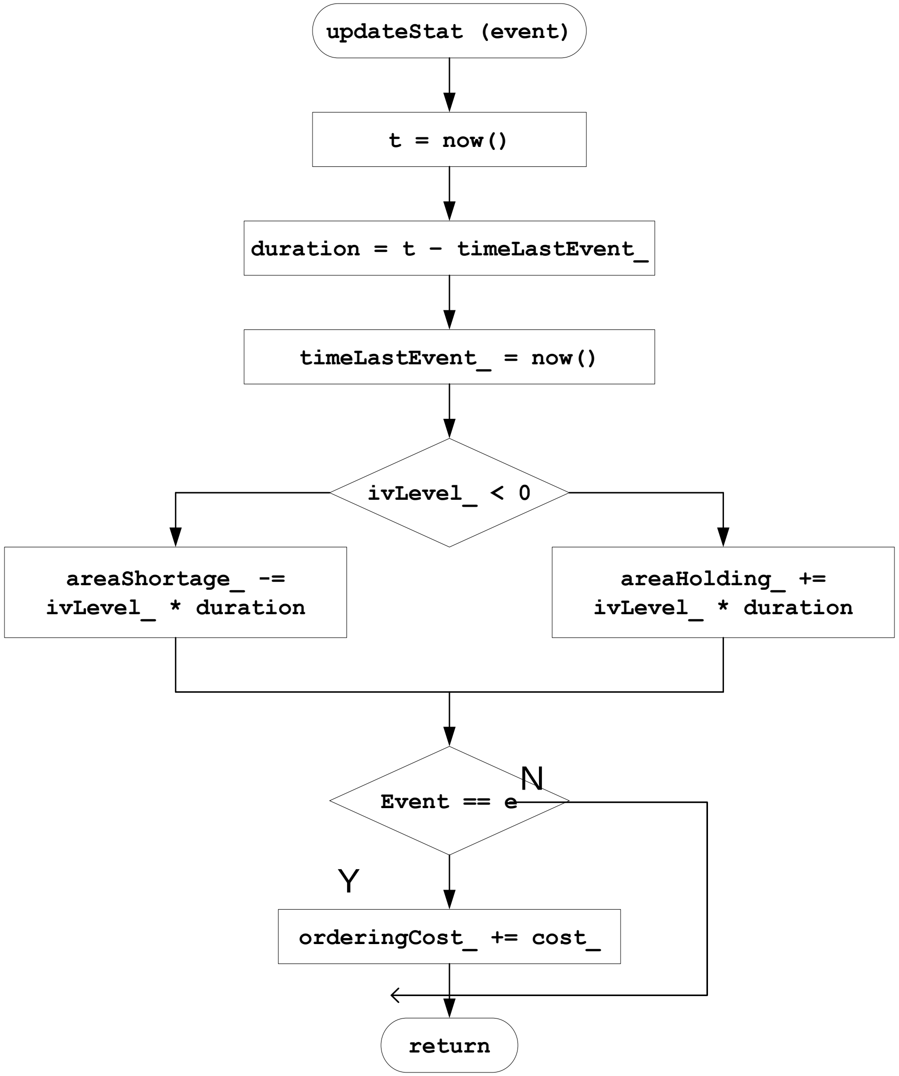
IS: Update Statistical Variables

avgOrderingCost = orderingCost_ / 120.0

avgHoldingCost = holdingCost_ * areaHolding_ / 120.0

avgShortageCost = shortageCost_ * areaShortage_ / 120.0;

avgCost = avgOrderingCost + avgShortageCost + avgHoldingCost



Different Policies

- $S_L = 20 \quad 20 \quad 20 \quad 20 \quad 40 \quad 40 \quad 40 \quad 60 \quad 60$
- $S_M = 40 \quad 60 \quad 80 \quad 100 \quad 60 \quad 80 \quad 100 \quad 80 \quad 100$